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# Class 9 · Maths · Weekly Test 1

**Sunday, 18 October 2026** · 60 minute · 40 marks · Test ID: T2026-10-19

Class 9 · Ch 1 Coordinates · Ch 2 Linear Polynomials · Ch 5 Circles (5.1–5.8)

Name: _____ Roll No.: _____

Date: _____ Score: _____ / 40

**Instructions:** Saare prashn compulsory hain. Section E mein internal choice hai. · Har step likho. Figure jahan zaroori ho, banao. · Calculator allowed nahi hai.

## Section A · MCQ + Assertion-Reason

**Q1 [1]** A point on the y-axis, 5 units below the origin, has coordinates: **1 mark**

a. (5, 0)
b. (–5, 0)
c. (0, –5)
d. (0, 5)

**Q2 [1]** The coordinates of every point on the x-axis are of the form: **1 mark**

a. (0, y)
b. (x, 0)
c. (x, x)
d. (x, y) with x = y

**Q3 [1]** Which of the following is a linear polynomial? **1 mark**

a. $x^2 + 1$
b. 5 – 4y
c. $x^3 - 2x$
d. 7

**Q4 [1]** The lines y = 2x + 3 and y = 2x – 5 are: **1 mark**

a. perpendicular
b. parallel
c. the same line
d. intersecting at (0, 3)

**Q5 [1]** Linear decay is represented by a straight line with: **1 mark**

a. positive slope
b. zero slope
c. negative slope
d. no y-intercept

**Q6 [1]** An arc subtends an angle of 70° at the centre of a circle. The angle it subtends at a point on the remaining part of the circle is: **1 mark**

a. 140°
b. 70°
c. 35°
d. 110°

**Q7 [1]** ABCD is a cyclic quadrilateral with ∠A = 75°. Then ∠C = **1 mark**

a. 75°
b. 105°
c. 115°
d. 15°

**Q8 [1]** The angle subtended by a diameter at any point on the circle is: **1 mark**

a. 45°
b. 60°
c. 90°
d. 180°

**Q9 [1]** How many circles can pass through three points that are not on one straight line? **1 mark**

a. None
b. Exactly one
c. Exactly two
d. Infinitely many

**Q10 [1]** **A:** Equal chords of a circle are at equal distances from the centre. **1 mark**
**R:** Of two unequal chords of a circle, the longer chord is closer to the centre.

a. Both A and R are true, and R is the correct explanation of A.
b. Both A and R are true, but R is not the correct explanation of A.
c. A is true, but R is false.
d. A is false, but R is true.

**Q11 [1]** **A:** The point (4, –4) lies on the line y = –x. **1 mark**

**R:** For every point on the line y = –x, the x-coordinate and the y-coordinate are equal.

a. Both A and R are true, and R is the correct explanation of A.

b. Both A and R are true, but R is not the correct explanation of A.

c. A is true, but R is false.

d. A is false, but R is true.

## Section B · Very Short Answer

**Q12 [2]** You have ₹800 and save ₹250 every month. Write the amount after n months as a linear pattern and find the amount after (i) 6 months (ii) 2 years. **2 marks**

**Q13 [2]** Find the distance of the point P(6, 8) from the origin. **2 marks**

**Q14 [2]** In a circle of radius 13 cm, a chord is 5 cm away from the centre. Find the length of the chord. **2 marks**

**Q15 [2]** PQRS is a cyclic quadrilateral with ∠P = (2x + 10)° and ∠R = (3x – 20)°. Find x, ∠P and ∠R. **2 marks**

## Section C · Short Answer

**Q16** [3] A positive number is 5 times another number. If 21 is added to both numbers, one of the new numbers becomes twice the other new number. Find the numbers. **3 marks**

**Q17** [3] M(–7, 1) is the midpoint of the segment joining A(3, –4) and B(x, y). Find the coordinates of B. **3 marks**

**Q18** [3] Two parallel chords of lengths 10 cm and 24 cm lie on the same side of the centre of a circle. The distance between them is 7 cm. Find the radius of the circle. **3 marks**

**Q19** [3] For the line y = –3x + 4: (i) write its slope and y-intercept, (ii) find the point where it cuts the x-axis, (iii) is this linear growth or linear decay? **3 marks**

## Section D · Case-Based

**Q20** **4 marks** **[4]**

A computer screen is 800 pixels wide and 600 pixels high, with the origin at the bottom-left corner. A circular icon of radius 80 pixels has its centre at A(100, 150). Another circular icon of radius 100 pixels has its centre at B(250, 230).

(i) Does any part of the circle with centre A go outside the screen? Justify. [1]
(ii) Does any part of the circle with centre B go outside the screen? Justify. [1]
(iii) Find the distance AB and decide whether the two circles intersect. [2]

## Section E · Long Answer

**Q21** **5 marks** **[5]**

Prove that the angle subtended by an arc at the centre of a circle is twice the angle subtended by it at any point on the remaining part of the circle. Draw a neat figure.

**OR**

A growing pattern of hexagons is made with matchsticks; each new hexagon shares one side with the previous one (Stage 1 = 1 hexagon, Stage 2 = 2 joined hexagons, ...). (i) Make a table of matchsticks for stages 1 to 4. (ii) Find the rule for stage n. (iii) How many matchsticks are needed for stage 15? (iv) Can 200 matchsticks form a stage exactly? Justify.

Student copy — solved answers are in the separate parent key.